

GROUP-FAIR STOCHASTIC GRADIENT DESCENT VIA MASKED OPTIMAL TRANSPORT

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ABSTRACT

Minibatch stochastic gradient descent (SGD) is group-blind: a batch contains each group in proportion to its prevalence, not to its weight in the objective, so SGD optimizes a prevalence-weighted surrogate. Reading a minibatch as a partial-participation event over groups, we carry the masked optimal transport construction of FedAVOT across: aligning the batch-composition law q with the target importance p yields per-batch group weights whose stochastic subgradient is exactly unbiased, at the $O(1/\sqrt{T})$ rate. Alignment is feasible under a Hall-type condition; past it, Sinkhorn returns an information projection with closed-form residual, splitting the excess risk into an instance-fixed alignment gain and an optimization residual paid by the estimator. A sweep over eleven availability skews on Adult and IMDB-Wiki shows the two crossing where the decomposition predicts. A mean-conditional-value-at-risk (CVaR) layer reaches the worst group at rate $O(\kappa(\alpha, \gamma)/\sqrt{T})$.

Index Terms— Group fairness, optimal transport, stochastic gradient descent, distribution shift, conditional value at risk.

1. INTRODUCTION

Imbalanced learning. In imbalanced classification and imbalanced regression the training data over-represent some groups and starve others: one demographic, one age range, one race [1, 2, 3, 4]. Call the groups one cares about *critical*. The objective is then not the empirical risk but $F_p(\theta) = \sum_i p_i f_i(\theta)$, with f_i the risk on group i and p the weighting one actually wants: uniform, inverse to prevalence, or externally dictated [5, 6].

What SGD does to a critical group. Minibatch SGD never sees p . A batch B^t of b i.i.d. examples contains group i in proportion to its prevalence π_i , so the group-blind update descends the prevalence-weighted risk:

$$\mathbb{E} \left[\frac{1}{b} \sum_{(x,y) \in B^t} \nabla \ell(m(x; \theta), y) \right] = \sum_i \pi_i \nabla f_i(\theta) = \nabla F_\pi(\theta). \quad (1)$$

The loss surface is tilted toward the represented groups by a bias, $\pi - p$, that survives $T \rightarrow \infty$ and every stepsize schedule (Sec. 2 generalizes (1) to any sampler).

The usual remedies. The standard response is to up-sample the rare groups or down-sample the common ones and then run SGD as usual. Tilting the sampler can only move the observation rate r_i of a group up to the batch’s coverage, so it cannot help a group with $p_i > r_i$; re-weighting the examples that do appear, the reweighting form of up-sampling, is a one-shot heuristic for a marginal problem

this paper solves exactly, and its un-normalized form, the fixed multiplier, destabilizes the iterates [7].

Other algorithms. Reduction methods for group fairness return randomized mixtures or carry an approximation constant tied to a fixed class [8, 9, 10, 11]; distributionally robust and worst-group objectives [3, 2, 12, 13, 14] change the functional rather than the sampler and are complementary; importance sampling for SGD [15, 16] corrects variance, not the identity of the minimized objective. The same mismatch is the central object of federated learning, where the *availability distribution* q over participating clients rarely matches the *importance distribution* p , so FedAvg [17] and its variants [18, 19] optimize a surrogate whose weights are induced rather than chosen [20, 21, 22]; FedAVOT [7] removes it by a masked optimal transport (MOT) problem on the client-round pairs that occurred. A minibatch is a participation event, its groups the active set with law q and target weighting p : group-fair SGD is federated aggregation on one machine. We call the method GAVOT; the aggregation rule is that of FedAVOT.

Contributions.

1. *The reduction* (Sec. 2). Group-blind minibatch SGD is agnostic FedAvg with one local step, so [20] names the function it minimizes, and transport weights make the batch subgradient exactly unbiased for ∇F_p (Lemma 5).
2. *A coverage law, a severity measure, and the crossover* (Corollary 4, Theorem 8, Sec. 5). Feasibility is Hall-type and reduces on singletons to $p_i \leq r_i$, so the *infeasible mass* $\nu \triangleq \sum_{i: p_i > r_i} p_i$ orders instances before training; past it the excess splits into an exact, instance-fixed *alignment gain* Δ_{flor} , the distance between the two optima the rules converge to, and an optimization residual paid by the estimator. We sweep eleven skews and watch the two cross.
3. *Risk-averse layer* (Sec. 4). Mean-CVaR over groups composes with the transport weights without disturbing unbiasedness, at rate $O(\kappa(\alpha, \gamma)/\sqrt{T})$, $\kappa(\alpha, \gamma) = \gamma + (1 - \gamma)/\alpha$.

2. PROBLEM FORMULATION

Groups and objectives. Let the population be partitioned into N groups indexed by $i \in [N]$ with *prevalences* $\pi \in \Delta^{N-1}$, $\pi_i = \mathbb{P}[\text{group } i]$. Group i carries a distribution $\mathcal{D}_i$ on $\mathcal{X} \times \mathcal{Y}$ and the risk $f_i(\theta) \triangleq \mathbb{E}_{(X,Y) \sim \mathcal{D}_i} [\ell(m(X; \theta), Y)]$, $\theta \in \mathcal{C} \subseteq \mathbb{R}^{d'}$, with $m(\cdot; \theta)$ the predictor and ℓ the loss. Empirical risk minimization weights groups by prevalence, $F_\pi = \sum_i \pi_i f_i$; the remedy is to fix an *importance distribution* $p \in \Delta^{N-1}$ and minimize

$$F_p(\theta) \triangleq \sum_{i=1}^N p_i f_i(\theta), \quad \theta^* \in \arg \min_{\theta \in \mathcal{C}} F_p(\theta), \quad (2)$$

uniform p being the category-uniform objective, $p_i \propto 1/\pi_i$ the inverse-prevalence one, or p a fairness constraint [5, 6].

Assumption 1. $\mathcal{C}$ is convex and compact with diameter D ; each f_i is convex on $\mathcal{C}$ and $0 \leq f_i \leq M$; every stochastic subgradient $g_i(\theta)$ of f_i satisfies $\mathbb{E}\|g_i(\theta)\|^2 \leq G^2$ on $\mathcal{C}$ (the assumptions of [20, 7] restated for groups; nothing restricts p or π).

Prevalence versus importance. Minibatch SGD does not see p . At step t the sampler draws a batch B^t of b examples; the *active set* $A^t \triangleq \{i : B^t \text{ contains an example from group } i\}$ has support $\{A_j\}_{j \in [M]}$ and *composition law* $q_j \triangleq \mathbb{P}[A^t = A_j]$. The observation rate of group i is $r_i \triangleq \mathbb{P}[i \in A^t] = \sum_{j: i \in A_j} q_j$, under i.i.d. sampling $1 - (1 - \pi_i)^b$: a group whose prevalence is below the resolution of the batch is absent from most batches whatever p says.

What group-blind SGD minimizes. The group-blind update gives group $i \in A^t$ the weight $\hat{\pi}_i^t$, its batch share, and absent groups zero; averaging over the batch given A^t and then over $A^t \sim q$ [20, Eq. 4] names the function SGD in fact minimizes.

Proposition 2 (Surrogate objective). *Under Assumption 1, group-blind minibatch SGD is a stochastic subgradient method on $F_{\tilde{p}}$ with*

$$\tilde{p}_i = \sum_{j: i \in A_j} q_j \mathbb{E}[\hat{\pi}_i^t | A^t = A_j], \quad (3)$$

a probability vector. For i.i.d. sampling $\tilde{p} = \pi$; for uniform weighting within the batch $\tilde{p}_i = \sum_{j: i \in A_j} q_j / |A_j|$, the surrogate of agnostic FedAvg [7, Eq. 2]. Consequently, for every $\theta \in \mathcal{C}$,

$$|F_p(\theta) - F_{\tilde{p}}(\theta)| \leq M\|p - \tilde{p}\|_1, \quad (4)$$

and any method that minimizes $F_{\tilde{p}}$ exactly incurs the excess $F_p(\theta_{\tilde{p}}^*) - F_p(\theta^*) \leq 2M\|p - \tilde{p}\|_1$ on (2), which does not vanish with T .

(Proof omitted for space.) The choice of p is thus decorative unless the aggregation step changes: the bias $p - \tilde{p}$ survives every stepsize schedule, rebalancing the sampler cannot help a group with $p_i > r_i$ since $\tilde{p}_i \leq r_i$ always. The question of [7], read on batches: *can the per-batch group weights be chosen so that SGD driven by q converges for the objective defined by p ?*

3. PROPOSED APPROACH: ALIGNMENT BY MASKED OPTIMAL TRANSPORT

Weights as a coupling. Let $Y[i, j] \geq 0$ be the weight of group i when the active set is A_j , $Y[i, j] = 0$ for $i \notin A_j$, and $T[i, j] \triangleq q_j Y[i, j]$ the joint mass carried from composition j to group i . Weights that form a probability vector on each realized composition and reproduce p in expectation are the *masked optimal transport* (MOT) feasibility problem of [7],

$$T \mathbf{1}_M = p, \quad \mathbf{1}_N^T T = q, \quad T \geq 0, \quad \text{supp}(T) \subseteq E, \quad (5)$$

with mask $E = \{(i, j) : i \in A_j\}$ (Fig. 1): an absent group cannot be weighted, and unlike classical transport [23, 24] there is no cost to minimize.

When alignment is possible. Existence is a supply–demand question on the bipartite graph E , settled by flow duality [25, 26] (proofs omitted for space).

Theorem 3 (Feasibility, [7, Thm. 1]). *For $I \subseteq [N]$ let $\mathcal{N}(I) = \{j : \exists i \in I, (i, j) \in E\}$ and $\mathcal{S}(I) = \{j : A_j \subseteq I\}$. Problem (5) is feasible if and only if*

$$\sum_{j \in \mathcal{S}(I)} q_j \leq \sum_{i \in I} p_i \leq \sum_{j \in \mathcal{N}(I)} q_j \quad \forall I \subseteq [N]. \quad (6)$$

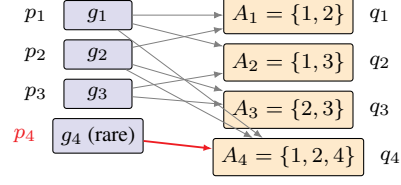


Fig. 1. The mask E : compositions carry the mass q they occur with, groups must receive the mass p the objective assigns. Group 4 is reachable only through A_4 , so alignment needs $p_4 \leq q_4$, the Hall condition of Theorem 3 on $I = \{4\}$; a smaller batch deletes A_4 and makes p_4 unreachable.

Algorithm 1 GAVOT: group-fair SGD by masked transport

Require: $p; (q, \{A_j\}, E); \eta; T; (\alpha, \gamma), \gamma = 1$ risk-neutral

1: $(T, Y) \leftarrow$ Sinkhorn scaling for (p, q, E) $\triangleright$ offline, once

2: $\theta^0 \in \mathcal{C}, t^0 \in \mathbb{R}$

3: **for** $t = 1, \dots, T$ **do**

4: draw batch B^t ; set A^t , let $j(t)$ be s.t. $A_{j(t)} = A^t$

5: $g_i^t \leftarrow$ stochastic subgradient of f_i from $B^t \cap$ group i

6: $w_i^t \leftarrow Y[i, j(t)](\gamma + \frac{1-\gamma}{\alpha} \mathbf{1}\{\hat{f}_i^t > t^{t-1}\})$

7: $\theta^t \leftarrow \Pi_{\mathcal{C}}(\theta^{t-1} - \eta \sum_{i \in A^t} w_i^t g_i^t)$

8: $\delta^t \leftarrow 1 - \frac{1}{\alpha} \sum_{i \in A^t} Y[i, j(t)] \mathbf{1}\{\hat{f}_i^t > t^{t-1}\}$

9: $t^t \leftarrow t^{t-1} - \eta(1 - \gamma) \delta^t$

10: **end for**

Ensure: $\bar{\theta}_T = \frac{1}{T} \sum_{t=1}^T \theta^t$

Corollary 4 (Coverage law and severity). *Feasibility of (5) requires $\mathbb{P}[A^t = \{i\}] \leq p_i \leq r_i$ for every i , with r_i the observation rate. Writing*

$$\nu \triangleq \sum_{i: p_i > r_i} p_i \quad (7)$$

for the infeasible mass, $\nu = 0$ iff the upper constraints hold. Under i.i.d. sampling of b examples the upper constraint reads $b \geq \log(1 - p_i) / \log(1 - \pi_i)$, under K -of- N group sampling $p_i \leq K/N$; ν is the axis Sec. 5 sweeps.

Computing the plan. Given feasibility, a coupling is obtained by Sinkhorn scaling on the masked support [27, 28, 24], $O(|E|)$ per sweep with M the number of *realized* compositions, once, offline. Since $\mathbf{1}_N^T T = q$ and $Y = T \text{diag}(q)^{-1}$, every column $Y[\cdot, j]$ is a probability vector on A_j , the property each bound uses. Algorithm 1 is the resulting method; $\gamma = 1$ is the risk-neutral rule analyzed here, $\gamma < 1$ the layer of Sec. 4.

Analysis. Let $\hat{g}(\theta) = \sum_{i \in A^t} Y[i, j(t)] g_i(\theta)$ be the transport-weighted batch subgradient of Algorithm 1. Everything rests on [7, Lem. 1] moved from models to gradients and on the classical projected-subgradient bound [29]: transport removes bias, it is not a variance reduction device.

Lemma 5 (Unbiasedness). *Let T satisfy (5). Then $\mathbb{E}[\hat{g}(\theta) | \theta] \in \partial F_p(\theta)$ for every $\theta \in \mathcal{C}$.*

Lemma 6 (Second moment, free of coverage). $\mathbb{E}\|\hat{g}(\theta)\|^2 \leq G^2$ for every $\theta \in \mathcal{C}$, with no dependence on $|A^t|$ or on N .

Theorem 7 (Rate). *Under Assumption 1, if (5) is feasible then Algorithm 1 with $\gamma = 1$ and $\eta = D/(G\sqrt{T})$ satisfies, with the constant of full-participation SGD and no dependence on batch coverage,*

$$\mathbb{E}[F_p(\bar{\theta}_T)] - F_p(\theta^*) \leq \frac{DG}{\sqrt{T}}. \quad (8)$$

Outside the feasible region. When (6) fails, Sinkhorn accumulates at the coupling whose row marginal $\hat{p} \triangleq T\mathbf{1}$ is Kullback–Leibler (KL) closest to p over the reachable polytope $\{T\mathbf{1} : T \geq 0, \mathbf{1}^\top T = q, \text{supp}(T) \subseteq E\}$ [30, 31, 24]. The column marginal is met exactly, so both lemmas hold with $\hat{p}$ for p : unbiased SGD on the surrogate $F_{\hat{p}}$, with $\hat{p}_i \leq r_i$ on singletons, so groups with $p_i > r_i$ are truncated at their observation rate.

Theorem 8 (Infeasible regime). *Let $\hat{p}$ be the row marginal of the Sinkhorn limit and $\theta^\dagger \in \arg \min_C F_{\hat{p}}$. For a weight vector w write $\Phi(w) \triangleq F_p(\arg \min_C F_w)$, the p -risk of the minimizer of the w -weighted risk. Then*

$$\mathbb{E}[F_p(\bar{\theta}_T)] - F_p(\theta^*) = \underbrace{\Phi(\bar{p}) - \Phi(p)}_{\text{alignment floor}} + \underbrace{\varepsilon_T(\bar{p})}_{\text{optimization residual}}, \quad (9)$$

where $\varepsilon_T(\bar{p}) \triangleq \mathbb{E}F_p(\bar{\theta}_T) - F_p(\theta^\dagger)$ satisfies $\mathbb{E}F_{\bar{p}}(\bar{\theta}_T) - F_{\bar{p}}(\theta^\dagger) \leq DG/\sqrt{T}$ and $\varepsilon_T(\bar{p}) \rightarrow 0$ whenever $\bar{\theta}_T \rightarrow \theta^\dagger$. Group-blind averaging obeys (9) with $\bar{p}$ in place of $\hat{p}$. Writing $\Delta_{\text{floor}} \triangleq \Phi(\bar{p}) - \Phi(\hat{p})$ for the alignment gain, transport improves on group-blind averaging exactly when $\Delta_{\text{floor}} > \varepsilon_T(\bar{p}) - \varepsilon_T(\hat{p})$, and every floor is bounded by $\Phi(w) - \Phi(p) \leq 2M\|p - w\|_1$.

Both terms are available before training, $\hat{p}, \bar{p}$ from (p, q, E) and Φ exactly on instances with a closed-form or convex frontier; the gain decides the sign, the residual is a stepsize artifact (Sec. 5).

4. A RISK-AVERSE LAYER OVER GROUPS

Exact alignment delivers the p -weighted risk, an average, which can hide a group. Where the requirement is on the worst group instead, we compose a mean–CVaR functional over groups on top of the transport weights [20, 32, 33]:

$$F_p^{\alpha, \gamma}(\theta) \triangleq \gamma F_p(\theta) + (1 - \gamma) \text{CVaR}_\alpha^p(f(\theta)), \quad \alpha, \gamma \in (0, 1], \quad (10)$$

where CVaR_α^p is the conditional value at risk of the group-loss vector under p . The Rockafellar–Uryasev representation turns (10) into a joint minimization over (θ, t) ,

$$\Phi(\theta, t) = \gamma \sum_i p_i f_i(\theta) + (1 - \gamma) \left(t + \frac{1}{\alpha} \sum_i p_i [f_i(\theta) - t]_+ \right), \quad (11)$$

jointly convex, with subgradients in both arguments p -weighted sums of per-group quantities, so Lemma 5 applies unchanged: transport fixes which distribution the expectation is under, CVaR what is averaged (lines 6, 8 and 9 of Algorithm 1).

Theorem 9 (Rate under mean–CVaR). *Under the hypotheses of Theorem 7, Algorithm 1 with $\eta = \Theta(1/\sqrt{T})$ satisfies $\mathbb{E}[F_p^{\alpha, \gamma}(\bar{\theta}_T)] - \min F_p^{\alpha, \gamma} = O(\kappa(\alpha, \gamma)/\sqrt{T})$ with $\kappa(\alpha, \gamma) = \gamma + (1 - \gamma)/\alpha$. In the infeasible regime the bound of Theorem 8 acquires the same factor on its statistical term, the bias term being unchanged.*

Both $\gamma = 1$ and $\alpha = 1$ recover Theorem 7 exactly; sending $\alpha \downarrow 0$ drives (10) to the worst group at the cost $\kappa(\alpha, \gamma) \rightarrow \infty$: the parameter prices robustness in the convergence constant, not in the limit.

Remark 10 (Relation to constrained learning). $\max_i f_i$ is the exact-penalty limit of $\min f_0$ s.t. $f_i \leq c_i$, a fixed price on the largest violation matching penalized and constrained values once it exceeds the least ℓ_1 norm on the dual-optimal set [34]. The $\alpha \downarrow 0$ edge of (10) is thus the scalarization a constrained specification selects, with κ its price in the rate.

Table 1. Headline cells, tail-500 over 5 seeds, decaying stepsize. *full*: every group every step; *upsamp.*: weights $\propto p_i/r_i$ within the batch; *LDS* [4]: label-distribution smoothing, regression only. The fixed multiplier $(N/K)p_i$ reaches 0.670 and 8504 on the two infeasible instances and diverges on the aligned ones.

Instance	GAVOT	blind avg.	upsamp.	LDS	full
Adult, $\nu=0.60$	0.2269	0.2479	0.2288	–	0.2073
Adult, $\nu=0$	0.2075	0.2077	0.2075	–	0.2073
IMDb, $\nu=0.31$	86.06	91.63	86.74	94.28	82.95
IMDb, $\nu=0$	84.28	86.37	84.40	89.43	82.95

5. EXPERIMENTS

Instances and protocol. *Adult* income classification [35]: logistic regression ($\eta=0.1$) on 100 race-homogeneous groups of 30 samples, groups per race $\propto$ prevalence (85/10/3/1/1), target uniform over the five races ($p_i = 0.2$ for a smallest-race group). *IMDb-Wiki* age regression [36]: linear regression (mean-squared error, MSE) on 128-d face embeddings ($\eta=10^{-2}$) over 100 identities, cubic target $p_i \propto (101 - i)^3$. Observation rates are swept: Adult $r \propto p^\beta$, $\beta \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$, from prevalence ($\nu=0.60$) to aligned ($\nu=0$); IMDb-Wiki $r_i \propto i^\beta$, $\beta \in \{0, \frac{1}{2}, 1, \frac{3}{2}, 2, 3\}$, from uniform ($\nu=0.31$) to the mirror image of p ($\nu=0.88$), plus aligned $r_i \propto 101 - i$ ($\nu=0$): eleven instances. Each step observes $K=3$ groups drawn $\propto r$, each taking $H=5$ local steps before the weighted average: *GAVOT*; *group-blind averaging*, uniform over the observed groups ($\bar{p}_i = r_i/K$); the *fixed multiplier* with weights $(N/K)p_i$ on the observed groups [7]; *full*, every group every step weighted by p . Step-size $\eta_t = \eta/(1 + t/1000)$, fixed before the comparison and shared by every rule (faster decays widen the IMDb-Wiki gap, never flip a winner), 4000 steps, 5 seeds, F_p over the last 500 steps. Floors are exact.

E1: the headline cells. Table 1 reports the four headline instances. Transport wins on three: Adult under prevalence availability, 0.2269 against 0.2479 ($t=45$), and both IMDb cells, 86.06 against 91.63 ($t=21$) and 84.28 against 86.37 ($t=13$); on Adult aligned the two coincide at the oracle value, as Proposition 2 predicts when $\bar{p} = p$. Self-normalized upsampling recovers most of the gain and transport the rest ($t=5.8$, 2.6 against it); downsampling never beats it, and LDS, which reweights by label rarity rather than group observation, trails even group-blind averaging. On the least represented group the margin over upsampling widens (77.0 against 80.0, 73.6 against 75.6 on IMDb-Wiki).

E2: the least represented group. Fig. 3 reads the same runs on the group the target exists for. On Adult, transport lowers the loss on race Other at every skew: 0.073 against 0.119 on the prevalence instance (full 0.031), 0.031 against 0.035 aligned, matching full. On IMDb-Wiki the top-importance identities show the crossover of Table 2 one step earlier: transport wins at $\nu \leq 0.70$ (77.0 against 86.4 at $\nu=0.31$, full 72.2; 73.6 against 79.9 aligned) and loses at the three most infeasible skews. The single most starved group (largest p_i/r_i) is Other itself on Adult and the top identity on IMDb-Wiki, 253 against 331 at $\nu=0.31$ (full 219).

E3: the crossover, and where the sign comes from. On Adult (Table 2, Fig. 2) the measured gain tracks Δ_{floor} within 4% at three of five skews and exceeds it at the two most infeasible (0.0210 against 0.0110, 0.0181 against 0.0144). On IMDb-Wiki the gain sits below the floor by a residual close to constant in β , mean 1.9, and the sign flips exactly where Δ_{floor} crosses that value, between $\beta=1.5$ (2.56, gain +0.58) and $\beta=2.0$ (1.62, −0.03): Theorem 8 at finite T . The

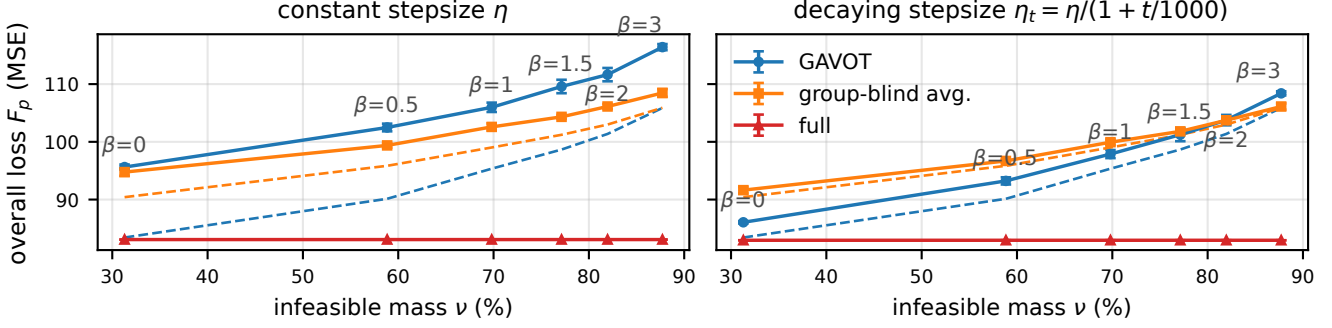


Fig. 2. IMDb-Wiki severity sweep: measured tail-500 (solid, ± 1 std) against the exact floor $\Phi(\cdot)$ of each rule (dashed). Left, constant stepsize: the residual swamps the alignment gain and transport losses everywhere. Right, decaying: the curves track their floors and transport wins until the floors close at high ν .

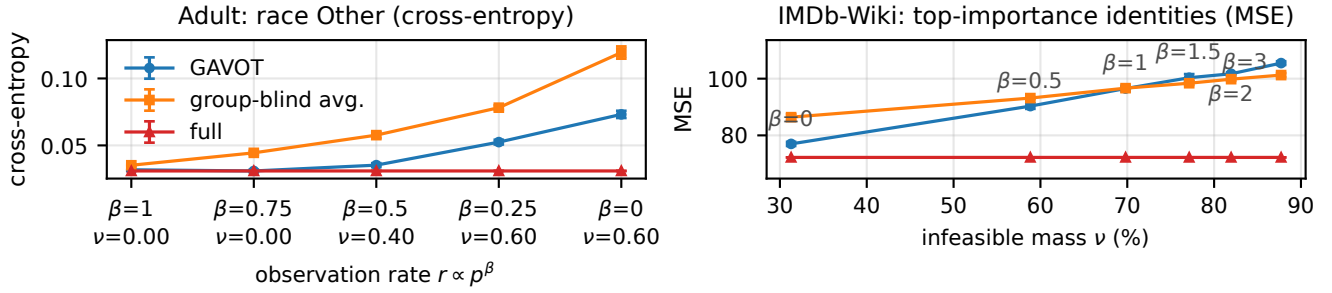


Fig. 3. Loss on the least represented group, decaying stepsize: Adult, cross-entropy on race Other ($p_i = 0.2$) against the observation exponent; IMDb-Wiki, MSE on the 20 top-importance identities, the least observed once $\beta > 0$. Same runs as Table 1.

Table 2. IMDb-Wiki severity sweep, decaying stepsize: Δ_{floor} the alignment gain of Theorem 8, *gain* the measured advantage of transport, t its Welch statistic. The residual averages 1.9; the sign flips where Δ_{floor} crosses it.

β	ν	Δ_{floor}	gain	t
0	0.31	6.99	+5.57	20.7
0.5	0.59	5.71	+3.46	11.3
1.0	0.70	3.67	+2.05	5.4
1.5	0.77	2.56	+0.58	1.1
2.0	0.82	1.62	−0.03	−0.1
3.0	0.88	0.02	−2.28	−7.9

ℓ_1 criterion cannot: $\|p - \hat{p}\|_1 < \|p - \tilde{p}\|_1$ in all eleven instances, the three where transport loses or ties included.

E4: why the sign flips. At constant stepsize (Fig. 2, left) transport loses on all six IMDb skews, by up to 7.9. A starved group has $p_i > r_i$, so $\hat{p}$ truncates at the observation rate; as ν grows $\hat{p}$ and $\tilde{p}$ starve the same groups and the floors converge (Δ_{floor} from 6.99 to 0.02, both 105.9 at $\nu=0.88$). Transport still concentrates weight on a starved group whenever it appears, so its variance about the mean is larger (Lemma 6 bounds only the second moment), and at constant stepsize the iterates sit on a noise floor proportional to it. Under $\eta_t = \eta/(1 + t/1000)$ that floor decays, the curves fall onto their dashed floors, and what remains is the residual, 2.3 at $\nu=0.88$, the one significant loss.

E5: risk aversion, and where it does not help. On a 10×10 (α, γ) grid the overall objective is minimized on the $\gamma = 1$ edge of every in-

stance: risk aversion costs the average. It earns its keep once, on the worst race of Adult under prevalence availability: GAVOT+CVaR at (0.2, 0.7) reaches 0.3624 against 0.3679 for GAVOT and 0.3994 for group-blind averaging. On the least represented group the layer never helps: the grid’s best rare-group loss is at $\gamma=1$ on all four headline instances. CVaR tilts among the groups present in the batch, and the rare group is absent from most; on top of transport it multiplies weights already p_i/r_i large and amplifies variance, not alignment.

6. CONCLUSION

Group-fair training fails at one fixable place: the aggregation step, blind to the weighting the objective declares. Reading a minibatch as a partial-participation event makes the fix the masked transport problem of federated learning. Transport is worth applying while the alignment gain exceeds the optimization residual, both computable before training, and the residual is a stepsize artifact a decaying schedule removes; the nonconvex case remains open.

7. REFERENCES

- [1] J. Buolamwini and T. Gebru, “Gender shades: Intersectional accuracy disparities in commercial gender classification,” in *Proc. Conf. Fairness, Accountability and Transparency (FAT*)*, 2018, pp. 77–91.
- [2] T. Hashimoto, M. Srivastava, H. Namkoong, and P. Liang, “Fairness without demographics in repeated loss minimization,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2018, pp. 1929–1938.
- [3] S. Sagawa, P. W. Koh, T. Hashimoto, and P. Liang, “Distributionally robust neural networks for group shifts,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2020.
- [4] Y. Yang, K. Zha, Y.-C. Chen, H. Wang, and D. Katabi, “Delving into deep imbalanced regression,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2021, pp. 11842–11851.
- [5] M. Mohri, G. Sivek, and A. T. Suresh, “Agnostic federated learning,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2019, pp. 4615–4625.
- [6] T. Li, M. Sanjabi, A. Beirami, and V. Smith, “Fair resource allocation in federated learning,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2020.
- [7] H. Rahimi and D. Kalogerias, “FedAVOT: Exact distribution alignment in federated learning via masked optimal transport,” in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2026.
- [8] A. Agarwal, A. Beygelzimer, M. Dudík, J. Langford, and H. Wallach, “A reductions approach to fair classification,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2018, pp. 60–69.
- [9] A. Cotter, H. Jiang, and K. Sridharan, “Two-player games for efficient non-convex constrained optimization,” in *Proc. Int. Conf. Algorithmic Learning Theory (ALT)*, 2019, pp. 300–332.
- [10] L. F. O. Chamon and A. Ribeiro, “Probably approximately correct constrained learning,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 16722–16735.
- [11] L. F. O. Chamon, S. Paternain, M. Calvo-Fullana, and A. Ribeiro, “Constrained learning with non-convex losses,” *IEEE Trans. Inf. Theory*, vol. 69, no. 3, pp. 1739–1760, 2023.
- [12] D. Levy, Y. Carmon, J. C. Duchi, and A. Sidford, “Large-scale methods for distributionally robust optimization,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 8847–8860.
- [13] H. Namkoong and J. C. Duchi, “Variance-based regularization with convex objectives,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 2971–2980.
- [14] S. Curi, K. Y. Levy, S. Jegelka, and A. Krause, “Adaptive sampling for stochastic risk-averse learning,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 1036–1047.
- [15] M. Rizk, S. Vlaski, and A. H. Sayed, “Optimal importance sampling for federated learning,” in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2021, pp. 3110–3114.
- [16] A. Katharopoulos and F. Fleuret, “Not all samples are created equal: Deep learning with importance sampling,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2018, pp. 2525–2534.
- [17] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Agüera y Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Proc. Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, 2017, pp. 1273–1282.
- [18] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, “Federated optimization in heterogeneous networks,” in *Proc. Machine Learning and Systems (MLSys)*, 2020, pp. 429–450.
- [19] S. P. Karimireddy, S. Kale, M. Mohri, S. J. Reddi, S. U. Stich, and A. T. Suresh, “SCAFFOLD: Stochastic controlled averaging for federated learning,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2020, pp. 5132–5143.
- [20] H. Rahimi and D. Kalogerias, “Convergence of agnostic federated averaging,” in *Proc. IEEE Int. Workshop Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP)*, 2025.
- [21] J. Wang, Q. Liu, H. Liang, G. Joshi, and H. V. Poor, “Tackling the objective inconsistency problem in heterogeneous federated optimization,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 7611–7623.
- [22] Y. J. Cho, J. Wang, and G. Joshi, “Towards understanding biased client selection in federated learning,” in *Proc. Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, 2022, pp. 10351–10375.
- [23] C. Villani, *Optimal Transport: Old and New*. Springer, 2009.
- [24] G. Peyré and M. Cuturi, *Computational Optimal Transport*. Now Publishers, 2019.
- [25] P. Hall, “On representatives of subsets,” *J. London Math. Soc.*, vol. 10, no. 1, pp. 26–30, 1935.
- [26] L. R. Ford and D. R. Fulkerson, “Maximal flow through a network,” *Canad. J. Math.*, vol. 8, pp. 399–404, 1956.
- [27] R. Sinkhorn, “Diagonal equivalence to matrices with prescribed row and column sums,” *Amer. Math. Monthly*, vol. 74, no. 4, pp. 402–405, 1967.
- [28] M. Cuturi, “Sinkhorn distances: Lightspeed computation of optimal transport,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 26, 2013, pp. 2292–2300.
- [29] A. Nemirovski, A. Juditsky, G. Lan, and A. Shapiro, “Robust stochastic approximation approach to stochastic programming,” *SIAM J. Optim.*, vol. 19, no. 4, pp. 1574–1609, 2009.
- [30] I. Csiszár, “ I -divergence geometry of probability distributions and minimization problems,” *Ann. Probab.*, vol. 3, no. 1, pp. 146–158, 1975.
- [31] C. Gietl and F. P. Reffel, “Accumulation points of the iterative proportional fitting procedure,” *Statistics*, vol. 47, no. 6, pp. 1321–1335, 2013.
- [32] R. T. Rockafellar and S. Uryasev, “Optimization of conditional value-at-risk,” *J. Risk*, vol. 2, no. 3, pp. 21–41, 2000.
- [33] P. Theodoropoulos, K. E. Nikolakakis, and D. Kalogerias, “Federated learning under restricted user availability,” in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2024.
- [34] H. Rahimi, S. Pougkakiotis, and D. Kalogerias, “An exact penalty method for constrained statistical learning over universal RKHSs,” submitted for publication, 2026.
- [35] B. Becker and R. Kohavi, “Adult,” UCI Machine Learning Repository, 1996. [Online]. Available: <https://doi.org/10.24432/C5XW20>
- [36] R. Rothe, R. Timofte, and L. Van Gool, “Deep expectation of real and apparent age from a single image without facial landmarks,” *Int. J. Comput. Vis.*, vol. 126, no. 2–4, pp. 144–157, 2018.